1. General

1.1. All dimensions are shown in millimetres unless noted otherwise. All levels shown are in metres unless noted otherwise. All dimensions, levels and setting out co-ordinates shall be checked on site. Any discrepancies shall be brought to the attention of the Designer's site representative.

1.2. Figured dimensions only are to be taken, do not scale from these drawings.

1.3. See other discipline drawings and Specifications for details of pavement, safety fences, drainage, earthworks etc.

1.4. This drawing is a copy of an electronic file and is copyrighted.

1.5. The structure is designed in accordance with Eurocodes.

1.6. This drawing shall be read in conjunction with the project Specification H3428-TYP-A-TI-SP-ST-0001 and the particular requirements which are shown on individual drawings.

1.7. All proprietary products shall be installed/applied in strict compliance with the Manufacturer's recommendations.

1.8. Any mention of the Contractor in the drawings refers to Dragados.

1.9. Temporary works will be submitted by the Contractor for Designer's review where permanent works affected.

1.10. Design loads are in accordance with the Approval in Principle H3428-TYP-A-TI-BD-ST-0001.

1.11 The Works shall be completed in accordance with the Specification For Highways Works (MCHW) as defined in the Contract. The Contract Specific Numbered Appendices relevant to this asset are included in the document H3428-TYP-A-TI-SP-ST-0001. The MCHW and associated Appendices are herein referred to as the 'Specification'.

1.12 The Contractor shall produce shop drawings and schedules as necessary for completion of the works and record drawings of the as-built works. All for the Designer's approval.

1.13. Consequence class for the whole structure is CC2. Inspection level for the whole structure is IL2.

2. Concrete

2.1. All concrete to be in accordance with BS EN 206:2013+A1:2016, BS EN 1992-1:2004 and BS EN 1992-2:2005 and the respectively National Annexes, BS8500-1 & BS8500-2 (+A2:2019) and the Specifications

2.2. All structures should achieve full 28-day strength before loading unless noted otherwise.

2.3. Where a formed surface is partially buried, the finish required on the exposed area shall continue to not less than 300mm below finished ground level.

2.4. All exposed arises shall have a 25x25mm chamfer unless stated otherwise. All kickers shall be 150mm high, unless otherwise stated.

2.5. The classes of concrete for the work, based on 28 days minimum compressive strength, according to AIP, are as follows:

Structural Element	Compressive Strength Class	Surface	Exposure Class				ΔC_{dev} (mm)	Nominal Cover C_{nom} (mm)
			XS	XC	XD	XF		
In-situ deck slab	C40/50	Top exposed surfaces	XS1	XC4	XD3	XF4	10	60
		Top surfaces covered by waterproofing	XS1	XC3	XD1	XF4	10	50
		Bottom and vertical surfaces	XS1	XC3	-	XF2	10	50
Abutment walls and wingwalls	C40/50	Exposed surfaces	XS1	XC4	XD2	XF1	10	55
		Buried surfaces	-	XC2	XD2	-	10	55
Pile caps	C32/40	Top and side surfaces	-	XC2	XD3	-	10	65
		Bottom surface	-	XC2	XD2	-	15	65
Piles	C32/40	Buried surfaces	-	XC2	XD2	-	50	105
Approach ramps retaining structures panels and copings	C40/50	All surfaces	XS1	XC4	XD3	XF1	10	60
Pier columns	C40/50	All surfaces	XS1	XC4	XD3	XF4	10	65
Deck diaphragms	P03 C40/50	Surfaces exposed to Robinson Road	XS1	XC4	XD3	XF4	10	65
		Other surfaces (including part of deck above diaphragms)	XS1	XC4	XD1	XF2	10	50
Pre-stressed beams	C50/60	Exposed faces	XS1	XC4	XD1	XF2	5	40
		Internal faces	-	XC3	-	-	5	30

2.5. Concrete mix design shall be as specified in Appendix 16/1 and 17/1 of the project Specificationsto satisfy durability requirements.

2.6. The required cover is to be maintained to any grooves and recesses on the concrete surface.

2.7. The Contractor shall take any necessary precautions to protect concrete surfaces from rust staining during construction. Any such surfaces stained during construction shall be cleaned to the satisfaction of the Designer's site representative.

2.8. All formed buried concrete surfaces shall be treated with two coats of bitumen paint as specified in the specification. Paint to extend up to no less than 150mm below proposed ground level.

2.9. Blinding concrete to be a minimum of 75mm thk, unless noted otherwise.

2.10. Waterproofing shall be in accordance with Appendix 20/1 of the project Specifications.

Concrete Finishes

Precast pre-stressed beams	F5 for exposed faces and F1 for hidden faces
In-situ deck soffit	F2
Unformed deck surfaces	U4
Deck slab	F4
Abutments, piers, diaphragms, wingwalls and approach ramps walls	F1, U1 for buried faces and F4, U3 for exposed faces
Piles (Unformed buried or concealed concrete surfaces)	U1
Surfaces to be waterproofed	F2, U4 or equivalent

2.11. In case GRP formwork is used, where the GRP ribs protrude into the concrete, the cover should be measured from the horizontal GRP/concrete interface, provided that at least 25 mm is also provided between the top of the rib and the nearest reinforcement, to allow rebar to be fully surrounded by concrete as per NG1704.4.

2.12. Anti-graffiti coating has not been included in the design

2.13. Construction joint to be provided on the concrete verges at not more than 6m longitudinal interval anda 20mm gap shall be allowed between consecutive pours.

2.14. Polymers shall not be used in the concrete mix of foundations and substructure.

3. Grout

3.1. Bedding grout shall have a minimum 28 days compressive strength of 50 N/mm²

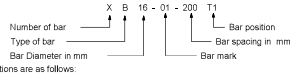
4. Reinforcement

4.1. Bars designated as "B" - high grade B500B or B500C steel shall conform to BS 4449:2005 and BS EN 10080:2005 +A2:2009.

4.2. Bars designated as "S" are stainless steel ribbed bars designation 1.4301 to BS 6744:2023 and strength grade 500.

4.3. All reinforcement to be cut or bent in accordance with BS 8666:2005.

4.4. Notation of bar reinforcement is as follows:



Bar positions are as follows:
- T: Top
- B: Bottom
- NF: Near Face
- FF: Far Face
- EF: Each Face
- SF: Bars in Side Face
- ABR: Alternate bars reversed
- ALT: Alternately placed
- ABS: Alternate bars staggered
- STG: Bars staggered

*Digit indicates Inner/Outer layer.

4.5. Lap length for unequal size bars shall be based on smaller bar.

4.6. Displace bars locally to avoid formed voids or obstructions less than or equal to 200mm in length or diameter with agreement of Designer's site representative.

4.7. Where reinforcement is detailed through openings, bars should be displaced locally on site. Bars should not be cut without the agreement of the Designer's site representative.

4.8. Welding of reinforcement (other than steel fabric, links to main bars in pile cages, lifting and stiffening bands) shall not be permitted.

4.9. Steel spacers of similar diameters to bars but not less than 25mm shall be provided between adjacent layers of parallel reinforcement and spaced at no more than 60 x the smallest diameter, if required.

4.10. Stated cover is nominal cover which includes a small fixing tolerance in accordance with BS 8500-1, BS EN 206-1, BS EN 1992-1-1 and National Annex to EN 1992-1-1
- +- 5 mm for precast produced in factory conditions
- +- 10 mm for insitu concrete
- +-15 mm for insitu concrete against blinding
- +- 50 mm for insitu piles

4.11. Minimum lap & anchorage lengths are given ontables below.

4.12. The Designer has considered guidance from the temporary works forum on stability of reinforcement cages as far as practicable in their permanent works design. The details shown on the reinforcement drawings do not include temporary works mitigation measures to ensure stability, strength and rigidity of reinforcement during lifting of prefabricated cages nor prior to, and during, concreting operations. The Contractor is responsible for the design and provision of temporary works such as chairs, spacers, bracing, clamps or external independent support to ensure stability, strength and rigidity of reinforcement during construction.

4.13. Good bond conditions generally apply to vertical reinforcement and at the bottom of slabs, diaphragms or copes. Poor bond conditions generally occur on the top of slabs, diaphragms or copes with a thickness greater than 250mm.

4.14. Where reinforcement couplers are specified, they shall be positioned to achieve the same nominal cover as that specified for the reinforcement bars that they couple.

4.15. The coupling system adopted must be certified as per project Specification. The manufacturer's installation instructions shall be strictly followed to ensure the full connection between bars is achieved. When specified, the coupling operations must be recorded during installation.

4.16. The reinforcement bar lengths for those that are to be connected by couplers, have been dimensioned in the bar bending schedules to extend to the centre of the coupling unit and with the two bars butting up against each other. The bar dimensions should be adjusted to suit the requirements of the coupler product used in construction.

4.17. The Contractor shall design and incorporate any additional reinforcement or temporary works required in order to support reinforcement as required.

4.18. Proposed anchorage and lap lengths values are indicated following tables:

Anchorage length Lbd in tension [mm]

Ø (mm)	C32/40				C40/50			
	Good bond		Poor bond		Good bond		Poor bond	
	Straight bars	Bars with standard hook*	Straight bars	Bars with standard hook*	Straight bars	Bars with standard hook*	Straight bars	Bars with standard hook*
8	275	275	395	395	240	240	340	340
10	345	345	490	490	295	295	425	425
12	415	415	590	590	355	355	510	510
16	550	550	785	785	475	475	675	675
20	685	685	980	980	590	590	845	845
25	855	855	1225	1225	740	740	1055	1055
32	1095	1095	1565	1565	945	945	1350	1350
40	1370	1370	1955	1955	1180	1180	1685	1685

Lap length Lo in tension [mm]

Ø (mm)	C32/40				C40/50			
	Good bond		Poor bond		Good bond		Poor bond	
	50% Lapped in one location	100% Lapped in one location	50% Lapped in one location	100% Lapped in one location	50% Lapped in one location	100% Lapped in one location	50% Lapped in one location	100% Lapped in one location
8	385	415	550	590	335	355	475	510
10	480	515	685	735	415	445	590	635
12	575	620	825	880	500	535	710	760
16	770	825	1095	1175	665	710	945	1015
20	960	1030	1370	1470	830	885	1180	1265
25	1200	1285	1710	1835	1035	1110	1475	1580
32	1535	1645	2190	2350	1325	1415	1890	2025
40	1920	2055	2740	2935	1655	1770	2360	2530

Anchorage length Lbd in compression [mm]

Ø (mm)	C32/40				C32/40			
	Good bond		Poor bond		Good bond		Poor bond	
	Straight bars	Bars with standard hook	Straight bars	Bars with standard hook	Straight bars	Bars with standard hook	Straight bars	Bars with standard hook
Ø	35-Ø	35-Ø	49-Ø	49-Ø	30-Ø	30-Ø	43-Ø	43-Ø

Lap length Lo in compression [mm]

Ø (mm)	C32/40				C40/50			
	Good bond		Poor bond		Good bond		Poor bond	
	50% Lapped in one location	100% Lapped in one location	50% Lapped in one location	100% Lapped in one location	50% Lapped in one location	100% Lapped in one location	50% Lapped in one location	100% Lapped in one location
Ø	48-Ø	52-Ø	69-Ø	74-Ø	42-Ø	45-Ø	59-Ø	64-Ø

5. Prestressing Strands

5.1. The prestressing strands shall be Y1860S7 low relaxation (class 2), stress-relieved seven wire superstrand complying with EN1992-2:2004 and as per EN 10308:2006.

5.2. The characteristic properties of the strands shall be:

Characteristic value of tensile strength, fpk = 1860MPa
Characteristic value of 0.1% proof stress, fp0.1k = 1640MPa
Modulus of Elasticity, E = 195,000MPa
Nominal diameter, d = 15.7mm
Nominal area, A = 150mm2
Nominal mass = 1.18kg/m
Maximum jacking force = 0.75 x fpk = 1395 MPa
Relaxation loss at 1000 hours under 0.7fpk = 2.5%

6. Fabrication and Erection

6.1. Lifting points shall be designed and detailed to suit the method of erection by the fabricator.

6.2. Drawings are certified for the permanent condition only. It is the fabricator's responsibility to verify structural components during all stages of erection. Any temporary works that affect the permanent works will require agreement with the permanent works Designer.

7. Existing site

7.1.Any uncharted buried services may be present. The location of buried services shall be confirmed and marked on site by the Contractor. Refer to drawings H3428-TYP-A-TI-DG-HT-0075 to 0082.

8. Setting out

8.1. The origin and orientation of the survey grid are to Ordnance Survey Of Great Britain 1936 (OSGB36) Grid System as defined within the topographical survey report.

8.2. All setting out coordinates are referenced to Ordnance Survey Of Great Britain 1936 (OSGB36) Grid System.

8.3. The Contractor shall check the setting out on site and report any discrepancies to the Designer prior to construction.

9. Geotechnical

9.1. Unless specified otherwise, the recommendations in the Specifications ground treatment drawings and Geotechnical Design Report H3428-TYP-A-XX-RP-CG-0001 shall be followed for the foundation preparation.

9.2. Groundwater control in the form of temporary pumping may be required within the excavation. Excavations should be kept free from water.

9.3. Fill to structures and retaining walls shall not be placed until the concrete has reached its 28 days strength.

9.4. Upon completion of excavation works, the Contractor shall consult with the Designer's site representative who will ensure that the ground condition at each structure location are in line with the detailed design considerations.

9.5. Foundations to be cast on blinding concrete.

10. Foundations and construction tolerances

10.1. Piles shall be installed with a tolerance of +/- 75mm on plan location unless otherwise specified.

10.2. The tolerance on the level of the reinforcement projecting above the final cut-off level shall be + 150mm/-50mm.

10.3. The tolerance on pile cut-off level shall be +/- 20mm.

10.4. Piles shall be installed with less than 1 in 75 deviation from vertical.

10.5. Refer to Appendix 16/1 of the project Specifications for piling information.

11. Health and Safety

11.1. Warning triangles (⚠) are provided to draw attention to particularly significant hazards or requirements. Such warnings shall be taken to apply to all similar features. The absence of these triangles does not imply that hazards do not exist elsewhere.

11.2. SHE boxes have been added to describe the residual risks in reference to the CDM warning triangle.

12. Pavement, Vehicle Restraint Systems and Fences

12.1. Vehicle restraint system to be H2 Class. Refer to Specifications for details.

12.2. The bridge perimeter will be secured as a permanent Restricted Area to ABP Physical Security Standard. A line of secure pedestrian parapet system will be installed onto the reinforced concrete upstand provided on each side of the bridge deck. This will be a proprietary product comprising a 2400mm high steel anti-climb pedestrian parapet system with additional mesh infill to maximise the anti-climb/security properties of the system.

12.3. Fence design is pending final approval by Contractor. Deck upstand reinforcement to be reviewed by the Designer once the fence anchoring details are confirmed.

12.4. For further details and information refer to drawings H3428-TYP-A-TI-DG-HT-0012, H3428-TYP-A-TI-DG-HT-0015, H3428-TYP-A-TI-DG-HT-0018, H3428-TYP-A-TI-DG-HT-0019, H3428-TYP-A-TI-DG-HT-0026, H3428-TYP-A-TI-DG-HT-0027, H3428-TYP-A-TI-DG-HT-0034 and H3428-TYP-A-TI-DG-HT-0085.

13. Precast Beams

13.1. The Contractor shall be responsible for designing a suitable method of lifting beams.

13.2. The Contractor shall be responsible for scheduling and dimensioning of all reinforcement within precast beams, in accordance with BS 8666:2020 the BS EN 15050:2007+A1:2012 and BS EN 13369:2013 Minimum reinforcement required in beams is shown in the respective beam drawings.

13.3. The beams stability during erection and construction shall be the responsibility of the Contractor. This is particularly important where beams are unsymmetrically loaded by additional deck edge elements and parapets.

13.4.Concrete of precast beams to reach a minimum of 40N/mm² strength before the transfer of prestressing force.

13.5. Contractor to check positioning of prestressed beams to ensure required alignment of projecting link bars and couplers is achieved prior to fixing deck reinforcement and casting slab.

13.6. Concrete cylinder strength at 28 days to be minimum 50N/mm², mean curing temperature 20°C, curing in accordance with EN 12390-2.

13.7. Top surface of the beam to be rough, a surface with at least 3mm roughness at about 40mm spacing, achieved by raking, exposing of aggregate or other methods giving an equivalent finish.

13.8. Ends of beams to be scabbled to form construction joints. Extent of scabbling shown on beam specific drawings.

14. Services

14.1. For details of ducts to be provided in bridge upstands, see utility drawings H3428-TYP-A-UT-DG-EL-0018 and H3428-TYP-A-UT-DG-EL-0019.

14.2. Ducts to have a minimum cover of 75mm.

14.3. The Contractor shall confirm the position of existing services prior to construction.

14.4 Lighting columns to be provided at intermediate pier locations. See lighting layout drawings for details. Deck reinforcement to be reviewed by the Designer once the lighting columns anchoring details are confirmed. Refer to drawings H3428-TYP-A-UT-DG-EL-0025 and H3428-TYP-A-UT-DG-EL-0026.

15. Drainage

15.1.Back of wall drainage will be provided at rear face of abutments. The drainage will be connected to the nearest drainage network beyond the abutments.

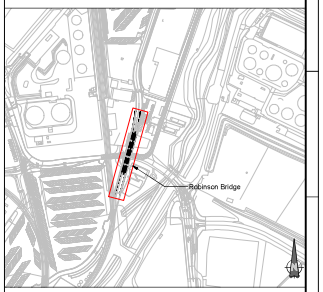
15.2. Combined kerb drainage units to be provided at each side of the carriageway for the full extent of the bridge and approach ramps. Refer to Specifications and drainage drawings for details. H3428-TYP-A-DI-DG-DR-0001 and from H3428-TYP-A-DI-DG-DR-0003 to H3428-TYP-A-DI-DG-DR-0016.

15.3. Gully details on the deck will be confirmed at construction stage.

16. Bearings

16.1. All bearings to be in accordance with the specification and as detailed on drawings. Fixing details and dimensions may change once a bearing Manufacturer/Designer has been appointed by the Contractor.

KEY PLAN



General Notes:

1. Coordinates in Ordnance Survey Of Great Britain 1936 (OSGB36) Grid System.
2. All dimensions in millimetres unless noted otherwise.
3. All levels to Ordnance Datum unless noted otherwise.
4. This design is based on the instructions received prior to issue and the assumptions included in the design package extract H3428-TYP-XX-XX-RG-XX-0009. All assumptions must be confirmed prior to proceeding to Construction on the basis of this design.

Rev.	Date	Description	By	Chk'd	App'd	ACC
C01	04/03/2026	ISSUED FOR CONSTRUCTION	LCT	LM	LSG	MF
P03	11/04/2025	DDL-5- Robinson bridge	LCT	LM	LSG	
P02	19/12/2024	DDL-5- Robinson bridge	LCT	LM	LSG	
P01	30/09/2024	DDL-5- Robinson bridge	LCT	LM	LSG	

Client: ABP ASSOCIATED BRITISH PORTS

Designer: DRAGADOS TYPSA

Project Title: H3428 - Immingham Eastern Ro-Ro Terminal

Drawing Title: ROBINSON BRIDGE GENERAL NOTES

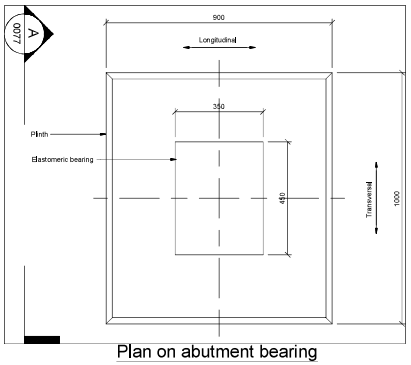
Date of Issue	Document Status	Drawn	Checked	Approved
04/03/2026	ISSUED FOR CONSTRUCTION	LCT	LM	LSG

Original Size	Sheet Number	Document Type	Suitability	Date	Date	Date
A1	1/1	DRAWING	A5	11/04/2025	11/04/2025	11/04/2025

Drawing Number: H3428-TYP-A-TI-DG-ST-0001

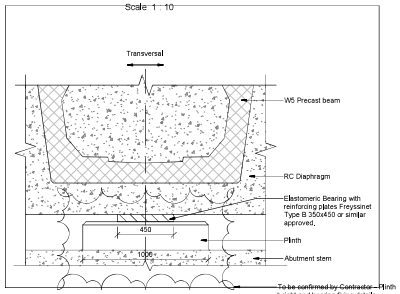
Revision: C01

Bearing identification mark				A0-1, A0-2, A0-3, A5-1, A5-2, A5-3
Symbolic representation in accordance with EN 1337-1:2000 Table 1				
Type of bearing (number in accordance with table 1)				1.1
Quantity				1
Seating material [1]	Material	Upper surface		In-situ concrete
		Lower surface		In-situ concrete
Support area	Characteristic strength (N/mm²)	Upper face		40
		Lower face		40
Actions Summary [kN] [2]	Serviceability Limit State (SLS) Maxima actions	Vertical	Max [kN] [2]	1688
			Permanent [kN]	748
			Min [kN] [2]	576
		Transverse Vy,Ed,ser [kN]		28
		Longitudinal Vx,Ed,ser [kN]		84
	Ultimate Limit State (ULS) Maxima Actions	Vertical [kN]		2261
		Transverse Vy,ud		37
		Longitudinal Vx,ud		121
	Accidental Limit State (ALS) Maxima Actions	Vertical [kN]	Max [kN]	1155
		Transverse Vy,Ed	Max [kN]	82
		Longitudinal Vx,Ed	Max [kN]	75
	Serviceability Limit State (SLS)	Permanent	Vertical NEd,ser [kN]	748
		Max Vertical Combination	Max Vertical NEd,ser [kN]	1688
			Coincident Transverse Vy,Ed,ser [kN]	2
			Coincident Longitudinal Vx,Ed,ser [kN]	65
			Coincident Displacement [mm]	8
			Coincident Rotation [rad]	3.51
		Min Vertical Combination	Min Vertical NEd,ser [kN]	576
			Coincident Transverse Vy,Ed,ser [kN]	24
			Coincident Longitudinal Vx,Ed,ser [kN]	25
			Coincident Displacement [mm]	10
			Coincident Rotation [mrad]	3.20
		Max Longitudinal Combination	Coincident Vertical NEd,ser [kN]	881
			Coincident Transverse Vy,Ed,ser [kN]	10
			Max Longitudinal Vx,Ed,ser [kN]	84
			Coincident Displacement [mm]	24
			Coincident Rotation [mrad]	3.94
	Ultimate Limit State (ULS)	Max Vertical Combination	Max Vertical NEd [kN]	2261
			Coincident Transverse Vy,Ed [kN]	3
			Coincident Longitudinal Vx,Ed [kN]	96
			Coincident Displacement [mm]	11
			Coincident Rotation [mrad]	4.85
		Min Vertical Combination	Min Vertical NEd [kN]	734
			Coincident Transverse Vy,Ed [kN]	35
			Coincident Longitudinal Vx,Ed [kN]	34
			Coincident Displacement [mm]	14
			Coincident Rotation [mrad]	4.22
		Max Longitudinal Combination	Coincident Vertical NEd [kN]	886
			Coincident Transverse Vy,Ed [kN]	13
			Max Longitudinal Vx,Ed [kN]	121
			Coincident Displacement [mm]	35
			Coincident Rotation [mrad]	5.24
	Accidental Limit State (ALS)	Max Vertical Combination	Max Vertical NEd [kN]	1155
			Coincident Transverse Vy,Ed [kN]	25
			Coincident Longitudinal Vx,Ed [kN]	22
			Coincident Displacement [mm]	16
			Coincident Rotation [mrad]	3.01
		Min Vertical Combination	Min Vertical NEd,ser [kN]	365
			Coincident Transverse Vy,Ed [kN]	81
			Coincident Longitudinal Vx,Ed [kN]	20
			Coincident Displacement [mm]	24
			Coincident Rotation [mrad]	2.96
		Max Transverse Combination	Coincident Vertical NEd [kN]	771
			Max Transverse Vy,Ed [kN]	82
			Coincident Longitudinal Vx,Ed [kN]	21
			Coincident Displacement [mm]	27
			Coincident Rotation [mrad]	2.97
Displacement [3] [7]	Serviceability Limit State (SLS)	Irreversible	Longitudinal Vx,sdi [mm]	6
			Transverse Vy,sdi [mm]	0
			Longitudinal αx,sdi [mrad]	1.07
		Reversible	Max Longitudinal Vx,Ed,ser,i [mm]	24
			Coincident Transverse Vy,Ed,ser,i [mm]	4
			Coincident Rotation [mrad]	3.94
			Coincident Vertical NEd,ser [kN]	697
			Coincident Transverse Vy,Ed,ser [kN]	3
			Coincident Longitudinal Vx,Ed,ser [kN]	9
	Ultimate Limit State (ULS)	Irreversible	Longitudinal Vx,sdi [mm]	6
			Transverse Vy,sdi [mm]	0
			Longitudinal αx,sdi [mrad]	1.07
		Reversible	Max Longitudinal Vx,Ed [mm]	35
			Coincident Transverse Vy,Ed [mm]	5
			Coincident Rotation [mrad]	5.24
			Coincident Vertical NEd [kN]	907
			Coincident Transverse Vy,Ed [kN]	5
			Coincident Longitudinal Vx,Ed [kN]	20
	Accidental Limit state (ALS)	Irreversible	Longitudinal vx,sdi [mm]	6
			Transverse vy,sdi [mm]	0
			Longitudinal αx,sdi [mrad]	1.07
		Reversible	Max Longitudinal Vx,Ed [mm]	14
			Coincident Transverse Vy,Ed [mm]	24
			Coincident Rotation [mrad]	2.97
			Coincident Vertical NEd [kN]	734
			Coincident Transverse Vy,Ed [kN]	22
			Coincident Longitudinal Vx,Ed [kN]	46
Rotation [mrad] [4] [5]	Serviceability Limit State (SLS)	Irreversible	Transverse αy,sdi [mrad]	0
			Longitudinal αx,sdi [mrad]	1.07
		Reversible	Max Longitudinal αx,Ed,ser,i [rad]	5.11
			Coincident Transverse αx,Ed,ser,i [mrad]	0.83
			Coincident Longitudinal Vx,Ed,ser,i [mm]	23
			Coincident Transverse Vy,Ed,ser,i [mm]	3
			Coincident Vertical NEd,ser [kN]	870
			Coincident Transverse Vy,Ed,ser [kN]	1
			Coincident Longitudinal Vx,Ed,ser [kN]	64
	Ultimate limit state (ULS)	Irreversible	Transverse αy,sdi [mrad]	0
			Longitudinal αx,sdi [mrad]	1.07
		Reversible	Max Longitudinal αx,Ed [mrad]	6.67
			Coincident Transverse αx,Ed [mrad]	1.13
			Coincident Longitudinal Vx,Ed [mm]	33
			Coincident Transverse Vy,Ed [mm]	5
			Coincident Vertical NEd [kN]	1150
			Coincident Transverse Vy,Ed [kN]	1
			Coincident Longitudinal Vx,Ed [kN]	95
	Maximum rate (100 x mrad / kN)		Transverse αy,sdm [mrad]	0.16
			Longitudinal αx,sdm [mrad]	0.59
Maximum bearing dimensions [mm]	Upper surface	Transverse		600
		Longitudinal		450
	Lower surface	Transverse		600
		Longitudinal		450
	Overall height			100
Type of fixing required	Upper face			Contractor's design
	Lower face			Contractor's design
Special requirements	No special requirements			



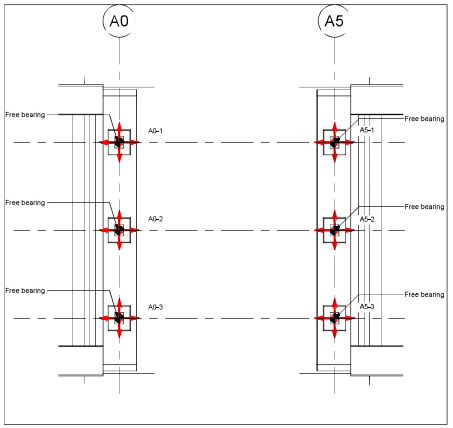
Plan on abutment bearing

Scale 1 : 10



A Section

Scale 1 : 20



Setting Out Plan

Scale 1 : 100

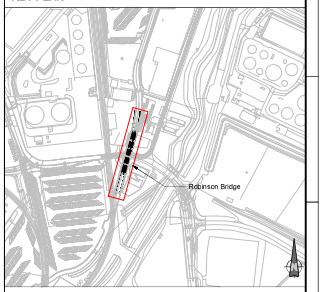
Specific notes:

1. Lower surface is provided by bearing plinths; upper surface is provided by deck diaphragms.
2. Maximum/Minimum load including permanent load.
3. Irreversible movement and rotation should be added to reversible values.
4. No minimum movement and rotation based on BS EN 1337-1:2000 Clause 5.5 were included.
5. Movement range determined considering bearing set temperature of 0°C for the expansion case and 20°C for the contraction case. Extra allowance of ±20°C was considered as per UK NA BS EN 1991-1-5:2003 Clause NA.2.6.
6. Positive vertical reaction indicates compression and negative vertical reaction indicates uplift.
7. Horizontal reactions obtained assuming horizontal stiffness equal to 3,457 kN/mm on all the bearings.

Bearing setting out points

Point	X	Y	Z
A0-1	520609,731	415446,187	6,941
A5-1	520588,134	415367,601	6,941
A0-2	520606,353	415447,116	6,941
A5-2	520584,756	415368,529	6,941
A0-3	520602,979	415448,041	6,941
A5-3	520581,381	415369,457	6,941

KEY PLAN



General Notes:

1. Coordinates in Ordnance Survey Of Great Britain 1936 (OSGB36) Grid System.
2. All dimensions in millimetres unless noted otherwise.
3. All levels to Ordnance Datum unless noted otherwise.
4. This design is based on the instructions received prior to issue and the assumptions included in the design package extract H3428-TYP-XX-XX-RG-X-0009. All assumptions must be confirmed prior to proceeding to Construction on the basis of this design.

Notes:

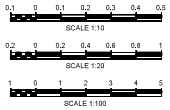
1. To be read in conjunction with drawings General Notes H3428-TYP-A-TI-DG-ST-0001 and General Arrangement H3428-TYP-A-TI-DG-ST-00020003
2. All typical and structure specific hazards to be identified on the contract drawings at detailed design stage. CDM Risk Register reference:H3428-TYP-C-XX-RA-RM-0001.
3. Bearing to be installed in accordance with manufacturer instruction.

Safety Health and Environmental information

In addition to the hazard and risks normally associated with the types of work detailed in this drawing, please note the following risks and information:

CDM risk	Phase	Hazard & description
1001	C	Potential for excessive foundation settlement resulting in overstressing of the proposed structure and/or unwanted cracking.
1013	C	Personal injury during lifting of large/heavy elements or falling loads.
1014	C	Potential difficult access/egress to the structure site and/or limited working space will endanger site operatives during transportation of structural elements, erection operations and vehicular manoeuvring.
1016	M	Low headroom at front of abutments.
1028	M	Health hazards associated with use of hazardous materials that requirement replacement or maintenance. Procedural error in the use of a toxic product for a corrective preventive maintenance operation or work. Risks of working with chemicals resulting in risk of injury to personnel.
1031	C	Health issues arising from excessive dust from post drilling, slip forming, in situ concrete pours etc.
1036	M	Risk of falling from height during inspection and replacement of bearings.
1042	M	Jacking operation during bearing replacement. Jacking of bridge leads to cracking of structure deck.
1043	M	Exposure to silica dust during maintenance activities for bearing replacement, joint replacement, parapet replacement following a collision leading to health issues.
1047	M	Loss of restraint to bridge deck during bearing replacement leading to excessive movement of the deck and/or overloading of adjacent bearings.

Phase: C = Construction, M = Maintenance, U = Use, D = Demolition



C01	04/03/2026	ISSUED FOR CONSTRUCTION	LCT	LM	LSG	MF
P03	11/04/2025	DDL-A- Robinson bridge	LCT	LM	LSG	
P02	19/12/2024	DDL-A- Robinson bridge	LCT	LM	LSG	
P01	30/09/2024	DDL-A- Robinson bridge	LCT	LM	LSG	
Rev.	Date	Description	By	Chk'd	App'd	ACC

Client: ABP ASSOCIATED BRITISH PORTS

Designer: DRAGADOS TYPSA

Project Title

H3428 - Immingham Eastern Ro-Ro Terminal

Drawing Title

ROBINSON BRIDGE
BEARING LAYOUT AND SCHEDULE

Date of Issue	Document Status	Drawn	Checked	Approved
04/03/2026	ISSUED FOR CONSTRUCTION	LCT	LM	LSG

Original Size	Sheet Number	Document Type	Suitability	Date	Date	Date
A1	1/1	DRAWING	A5	11/04/2025	11/04/2025	11/04/2025

Drawing Number: H3428-TYP-A-TI-DG-ST-0077

Revision: C01